

Yasasvi Nellore

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INTRODUCTION

Experienced software engineer with a strong background in designing scalable systems and delivering impactful solutions to enhance performance and user satisfaction. Skilled at streamlining processes, improving system reliability, and implementing effective integrations to meet business goals. Known for collaborating closely with teams to develop innovative features and ensure the successful delivery of projects. Committed to building efficient, reliable applications that drive meaningful results.

SKILLS

Languages: Java, C, C++, Python, JavaScript, TypeScript, HTML/CSS, SQL, NoSQL, Scala, C#.

Frameworks and Libraries: Spring Framework (Spring Boot, Spring MVC, Spring Security) Hibernate, Flutter, React, Angular, Node.js, Express, Nest.js, NextJS, MarkoJS, Vue.js, Django.

Web Component APIs and Resource APIs: Servlets 3.1, JSP 2.3, JSF, JDBC and Web services (JAX-WS, JAX-RS), RESTful APIs, SOAP APIs

Databases: SQL Server, MySQL, Oracle Database, PL/SQL, MongoDB, HBase, Cassandra, Postgres

Big Data and Data Processing: Apache Spark (Spark SQL, Spark Streaming, PySpark), Kafka, Hive, Pig, Pandas

DevOps and Automation: Jenkins, Maven, Docker, Kubernetes, Node Package Manager (NPM), CI/CD Pipelines

Tools and Platforms: JIRA, Eclipse, IntelliJ IDEA, NetBeans, Visual Studio, JRebel, RabbitMQ, JBoss, BEA

WebLogic, WebSphere, Apache Tomcat, Bitbucket/Git, SVN

Web Technologies: HTML5, CSS3, Bootstrap, SASS, jQuery, AJAX, JSON

Cloud Platforms: AWS, Azure, GCP

Security and Testing: Single Sign-On (SSO), Data Encryption/Decryption, JUnit, Unit Testing

EXPERIENCE

Sr Backend Engineer, Data

Aug 2024 - Dec 2024

FlowAI

Texas, USA

- Developed and deployed AI-driven solutions to identify and deliver potential customer contacts, enhancing product sales and engagement. Spearheaded the implementation of a distributed microservices architecture .
- Configured Apache Spark Streaming for real-time data processing from Kafka to HDFS, designed ETL pipelines for seamless data integration from sources like Salesforce and SQL Server, and built scalable batch and real-time workflows using PySpark, Spark SQL, and Scala integrated with HBase.
- Developed and integrated RESTful and SOAP web services, leveraging Spring Framework for efficient back-end functionality and implementing Servlets and JSPs based on the MVC pattern.
- Designed optimized database schemas for high-traffic applications, reducing storage overhead through indexing and normalization. Worked extensively with NoSQL (MongoDB) and relational databases (Oracle 11g, SQL Server), writing advanced queries and triggers, and collaborated on seamless data migration from monolithic databases to microservices.
- Designed and developed integrations with Gmail, LinkedIn, Salesforce, and sequencing tools, driving increased user engagement, while enhancing system reliability and scalability with secure authentication flows and persistent user data storage.
- Created Python-based microservices for a platform with FastAPI, utilizing Pandas for data processing and transformation from various sources, including Excel and HBase.
- Worked extensively with Spring modules, including Spring Boot, Spring MVC, Spring Security, and Spring Integration, to build and deploy scalable applications.
- Managed message-driven architectures using RabbitMQ and JMS for seamless communication between distributed applications.
- Designed and deployed Docker containers and orchestrated them using Kubernetes, implemented continuous integration and deployment processes using Maven, Jenkins, and Git, ensuring smooth production releases and code versioning.
- Configured AWS services such as EC2 instances, VPCs, and S3 storage for deployment and maintenance of cloud-based applications.

Software Engineer

University of Texas at Arlington

Dec 2023 - May 2024

Texas, USA

- Developed a web-based application to manage inventory, track sales, and handle payment processing for multiple university food courts, including Marketplace, Subway, Panda Express and Connections. The system ensures real-time inventory updates, accurate sales tracking, and secure payment handling, tailored to meet the needs of a dynamic university environment.
- Designed a responsive Single Page Application (SPA) using React with user-friendly dashboards and forms utilizing JavaScript, TypeScript, and AJAX for dynamic functionality, implemented client-side validations with jQuery, and developed cross-platform desktop applications using Electron.js for seamless food court operations.
- Developed backend logic in Node.js, creating modules across multiple food courts. Built and deployed RESTful APIs with Express.js for seamless frontend-backend communication, implemented middleware for secure API requests, role-based access control, and logging, and integrated third-party libraries like Axios for real-time data synchronization.
- Designed interactive dashboards with React and Chart.js for administrators to monitor real-time sales, inventory levels, and payment statuses. Created SQL views and aggregation queries to provide analytics on sales performance, peak hours, and best-selling products.
- Streamlined development with Node Package Manager (NPM) for managing dependencies, automated builds and deployments using Jenkins, and deployed the application on AWS EC2 instances and Docker containers, ensuring high availability and scalability for university-wide operations.
- Led the development of a reusable Node.js and JSON Web Token (JWT) framework for Single Sign-On (SSO) to secure user authentication for administrators and food court staff.

Java Developer

Red Cloud Technologies

Aug 2021 - May 2022

Hyderabad, India

- A comprehensive web-based application designed to manage employee details, attendance, payroll, and performance reviews. This system automates routine HR tasks, ensuring efficient data management and seamless user interactions.
- Developed a responsive Single Page Application (SPA) using AngularJS, incorporating user controls and web forms for managing employee data, attendance tracking, and performance reviews, with JavaScript and jQuery enabling seamless client-side validations to ensure consistent and accurate data entry.
- Designed and implemented backend business logic using Core Java, developing modules for payroll, attendance, and performance management, along with base and utility classes featuring customized exception handling for robust error management, and built REST APIs to enable seamless communication between frontend and backend components.
- Designed and maintained relational database schemas using SQL for employee, attendance, and payroll data, optimizing queries for performance, developing stored procedures for payroll and attendance automation.
- Automated payroll generation and performance review notifications using SQL triggers and scheduled tasks, developed interactive dashboards for actionable insights, and automated the distribution of critical reports via email.
- Streamlined development with NPM for dependency management and applied modular design principles to improve code maintainability and scalability.
- Conducted end-to-end testing of frontend and backend workflows for reliability, logged and resolved defects using JIRA to ensure efficient issue resolution and quality control.

PROJECTS

Chess Bellator | *Python, Machine learning algorithms, PyTorch, NumPy*

Mar 2024

- Built and analyzed chess engines using Alpha-Beta Pruning and Monte Carlo Tree Search (MCTS) algorithms. Integrated CNNs to assess chess board positions, recognizing piece arrangements and adhering to chess rules. Trained a Convolutional Neural Network (CNN) model using **Python, PyTorch, and NumPy** on **50M** LiChess game records, extracting features from FEN strings to achieve **90%** evaluation accuracy.

Diffusion model to generate faces | *Huggingface- UNet2D- MetFaces*

Mar 2023

- Created a diffusion model using the Huggingface Diffusers library. Trained a UNet2D model from scratch on the MetFaces dataset using Google Colab Pro to generate human faces resembling works of art.

Student Academic Portal | *React-LaraveL-PHP-AWS*

May 2023

- Developed a user-friendly academic portal for university students using React.js for the front end and Laravel (PHP) for the back end. Implemented features for student progress tracking, real-time updates, and user authentication. Managed the implementation of the application on AWS, focusing on scalability, reliability, and cloud infrastructure management, ensuring seamless access and performance for a large user base.

- Currency Detector | *OpenCV- TensorFlow- AndroidStudio*

Dec 2023

 - Developed a Currency Detector app for visually impaired users, enabling currency recognition through voice commands and audio feedback. Utilized OpenCV for image processing and TensorFlow for training machine learning models to identify currency notes. Integrated Google Speech-to-Text for voice command recognition and Google Text-to-Speech for audio feedback. Designed the app using Android Studio to ensure accessibility and usability, collaborating with users during testing to gather feedback and enhance functionality.
- Automatic Job applications | *Python, OpenAI GPT, Selenium, OpenCV, Pandas, Docker*

May 2024

 - Developed a system using LLMs and WebDriver to automate job applications. The system dynamically parses HTML and screenshots to identify input fields, maps them to predefined user data, and fills out forms accurately. Leveraged Python, Selenium, OpenAI GPT, and BeautifulSoup for web automation and field validation, reducing manual effort and improving application accuracy.
- Shopping App | *MERN stack, MongoDB, Express, React, Node.js*

Oct 2023

 - Crafted an intuitively navigable e-commerce web application featuring a well-designed user interface (UI). Devised two distinct JWT-based authentication models tailored for guest and registered users. Developed the frontend with **ReactJS**, supported by a robust backend featuring over **20** APIs implemented with **Express** and **MongoDB**.

EDUCATION

The University of Texas, Arlington	Aug 2022 - May 2024
<i>Master of Science in Computer Science CGPA : 3.7/4</i>	<i>Arlington, TX</i>
GITAM University	
<i>Bachelor of Engineering in Computer Science and Engineering CGPA : 8/10</i>	<i>Hyderabad, India</i>

CERTIFICATIONS

- AWS Solution Architect Associate